

Ideation Phase

Define the Problem Statements

Attribute	Project Information
Project Title	Auto Ticket Classification using Flow Designer
Phase / Stage	1. Ideation Phase
Document Title	Define the Problem Statements
Platform & Scope	ServiceNow Utah / Vancouver / Washington DC (Global Scope)
Team ID	SWTID-2026-4988

1. Overview of Problem Statement Formulation

In educational institutions and campus environments, the IT helpdesk is the vital operational backbone ensuring uninterrupted learning and administrative productivity. Students and teachers encounter daily technical bottlenecks ranging from Wi-Fi dropouts and projector malfunctions to account lockouts and workstation freezing. Currently, ticket triage in many schools is performed manually, where IT staff must inspect each submitted ticket, interpret free-form text, and select appropriate category and subcategory classifications. This document articulates the specific customer problem statements through empathy-driven stakeholder lenses to guide the automated solution.

2. Customer Problem Statement Matrix

The following problem statements capture the perspective of various stakeholders facing challenges in the current incident management lifecycle:

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1 (Helpdesk Staff)	School IT Helpdesk Specialist / Agent	Triage and resolve incoming technical requests promptly and route them to appropriate hardware or network specialists.	I spend up to 40% of my workday manually reading raw tickets and picking Category and Subcategory options from dropdowns.	Incoming tickets submitted by students and faculty lack automated categorization at the time of record creation.	Overburdened by repetitive administrative overhead, exhausted during morning rush hours, and delayed in fixing urgent classroom outages.

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-2 (End User / Teacher)	High School Teacher / Faculty Member	Conduct smooth digital lectures using classroom projectors, smart boards, and campus Wi-Fi.	When hardware or network fails, my submitted support ticket sits unassigned and unacknowledged for hours.	The helpdesk team is overloaded with manual ticket sorting and lacks an instant automated notification mechanism.	Frustrated, helpless in front of students, and hindered in delivering planned curriculum and assessments.
PS-3 (Student Caller)	Enrolled Student	Access university learning management portals and submit homework assignments before deadlines.	When I experience password lockouts or laptop slowness, I submit a request but have no visibility into whether IT received it.	No immediate confirmation email is dispatched, and manual category selection delays queue assignment.	Anxious about missing assignment submission deadlines and disconnected from academic resources.
PS-4 (IT Operations Lead)	Campus IT Director / Support Manager	Enforce strict Service Level Agreements (SLAs), generate accurate ITSM analytics, and maintain audit-compliant incident records.	Categorization reports show high human error rates, misrouted tickets, and inconsistent categorization across shifts.	Triage relies solely on human discretion without standardized, automated keyword evaluation rules.	Concerned about distorted reporting metrics, SLA breaches, and suboptimal technician resource allocation.

3. Root Cause Analysis (5-Whys Methodology)

To ensure the proposed solution addresses root systemic vulnerabilities rather than mere surface symptoms, the 5-Whys diagnostic technique was conducted:

Level	Diagnostic Question	Root Cause Finding
Why 1	Why are classroom IT issues taking hours to receive first-touch routing?	Tickets sit in the general triage queue waiting for an IT dispatcher to categorize them.
Why 2	Why must tickets wait for manual dispatcher triage?	Incident records are submitted without predefined Category and Subcategory values.
Why 3	Why are callers not selecting the categories during submission?	Callers are students and teachers who find technical taxonomy confusing, leading to erroneous or abandoned classifications.

Level	Diagnostic Question	Root Cause Finding
Why 4	Why is there no automated system classifying tickets on submission?	The legacy ServiceNow implementation relied entirely on standard manual form submission without automated flow logic.
Why 5	What is the fundamental root cause?	Absence of an event-driven, keyword-based Flow Designer automation that analyzes ticket text and applies dependent taxonomy upon record insertion.

4. Measurable Goals and Success Criteria

1. Zero-Touch Triage: Automatically classify 100% of tickets containing predefined issue keywords upon record creation.
2. Immediate Caller Acknowledgment: Dispatch an automated confirmation email to the caller within 2 seconds of ticket submission.
3. Data Integrity: Enforce dependent choice relationships so Subcategories strictly align with their parent Categories.
4. Low-Code Sustainability: Implement the entire solution natively in ServiceNow Flow Designer without complex JavaScript scripting.